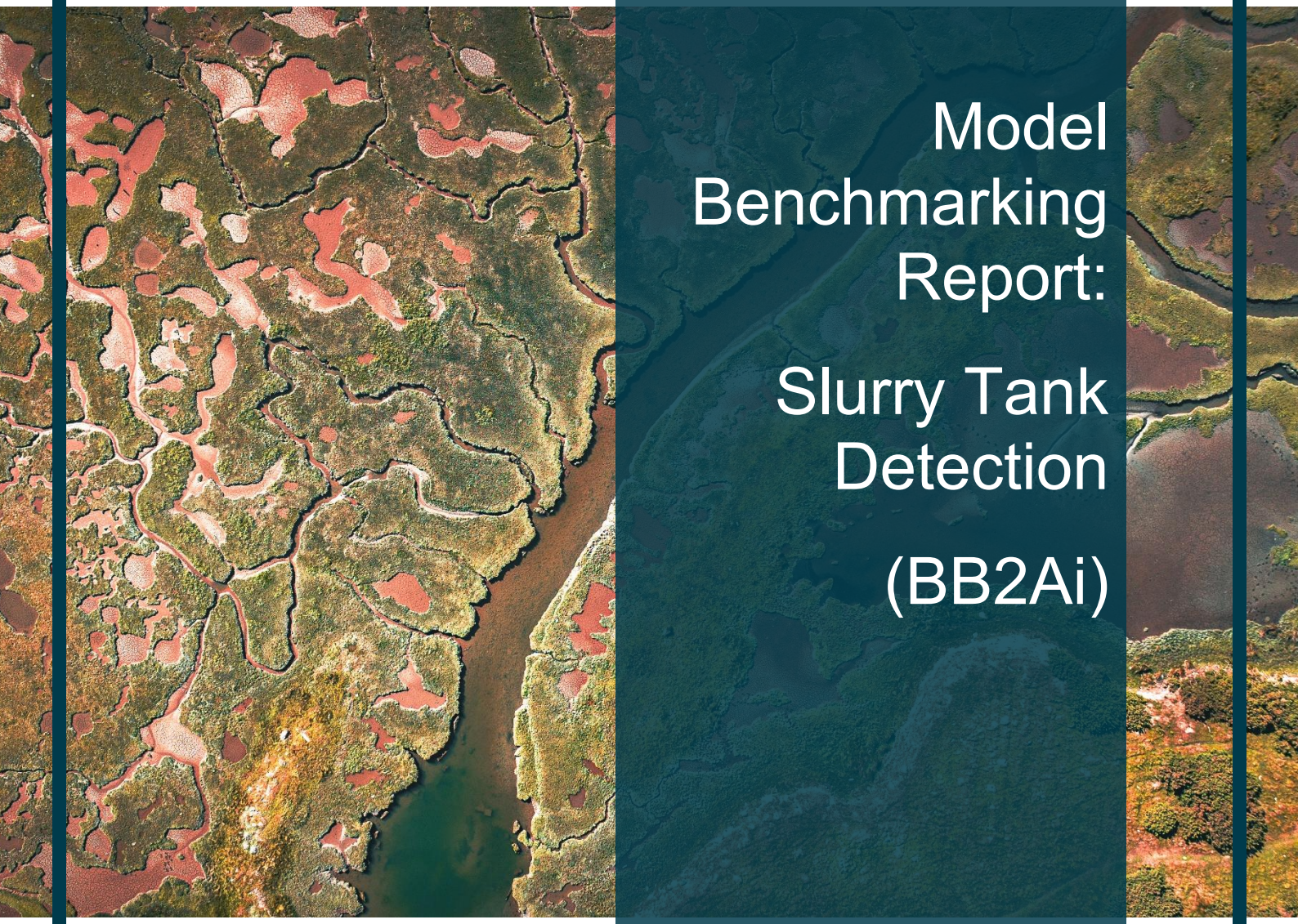
River Deep Mountain AI



Model Benchmarking Report:

Slurry Tank Detection

(BB2Ai)

ADAS

February 2026

Model Benchmarking Report - Slurry Tanks (BB2Ai)

FINAL REPORT

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Authors: Richard Gooday, Lionel Delaney, Pascoe Midgley, Harry Tricklebank, Aled Thomas.

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Glossary

ALERT	Agricultural Land Environmental Risk and Opportunity Tool
MAGIC	Multi-Agency Geographic Information for the Countryside
RDMAI	River Deep Mountain AI
WWTW	Waste Water Treatment Works

Summary

Table 1.1 Benchmarking summary

Model Outputs	Manual review of data / catchment walkover	RDMAI Model
Model Approach	Manual review of available datasets and in-person catchment walkover to identify features of interest.	The RDMAI Slurry Tank Detection model uses a YOLOv8 deep learning pipeline trained on public and custom data to determine the locations of above ground circular slurry tanks from detailed satellite imagery.
Applications	Identify areas of high risk for agricultural pollution and where placement of mitigation measures would be beneficial.	The model would produce an additional dataset to support catchment management and walkovers to help identify locations for mitigation of diffuse agricultural pollution.
Geographic Constraints	Theoretically none, although land access is an issue. Not all potential areas of interest are visible from roads, footpaths and other publicly accessible spaces.	The model was trained primarily on UK imagery. It should run anywhere with satellite imagery of similar spatial resolution and spectral characteristics (RGB bands, 0.3–1 m pixel resolution).
Temporal Resolution	Catchment walkovers can be undertaken as frequently as desired (given practical and cost considerations). However, slurry tanks are relatively permanent structures (typically with a minimum 20-year designed lifespan), so temporal resolution is not a concern.	The model does not have a built-in temporal component as it runs on the imagery provided. The temporal resolution is thus limited by the frequency of data provision and model application. However, slurry tanks are relatively permanent structures (typically with a minimum 20 year designed lifespan), so temporal resolution is not a concern.

Data Inputs Required	A manual search for slurry tanks can be undertaken using imagery data freely available on the internet such as that included in Google Maps.	Requires high-resolution (sub-1m) satellite or aerial imagery in GeoTIFF format. Images must be georeferenced.
Model Performance	One of the slurry tanks in the Welsh catchment was not identified as part of the walkover. Performance is user-subjective and subject to human error.	Insufficient data to draw a conclusion. In the Welsh catchment, the model found one tank, missed one tank and mis-identified a third site. More data is needed to decide the threshold for categorising the identified features as slurry tanks.
Cost & Time	A thorough review of multiple datasets for a small (~ 10 km^2) catchment could take a few hours, a fairly rapid catchment walkover of a similar area would take around a day. The only significant costs are staff time (~1 day) to undertake the work, which might be repeated annually to see if changes are being made.	Model run time is quick (c. 1 hour for a 40 km^2 catchment). The main constraint is the need to purchase detailed satellite imagery (approximately £10 per km^2), which means that the model is not particularly cost-efficient if the model is the only reason to purchase the data.
Strengths	A walkover provides more detailed insight into condition of the slurry tank and management of neighbouring farm land.	Relatively simple to use. Large-scale application is possible.
Limitations	A catchment walkover is relatively labour intensive. Performance is user-subjective and subject to human error.	Detailed satellite imagery required. The existing Github file does not include the 'model weights' file necessary to run the model, as licensing restrictions on the data used in the RDMAI project mean it cannot be shared. Therefore, this file needs to be created by the user. Above ground circular tanks are not the only option for storage of manure.

1. Benchmarking Methodology

1.1 Model Selection

1.1.1 RDMAI Slurry Tank Detection Model

Understanding the location of agricultural slurry tanks is important for assessing agricultural sources of pollution. First, they act as an indicator of intensive livestock production and the pollution risks associated with livestock grazing and manure application. Second, poorly maintained slurry tanks can leak nutrients and organic matter into nearby surface water and groundwater. Therefore, slurry tanks are indicators of both potential point and diffuse sources of agricultural pollution.

The Slurry Tank Detection Model identifies agricultural circular above-ground slurry storage tanks using high-resolution satellite imagery. Using a YOLOv8 deep learning pipeline trained on public and custom datasets, the model can generate georeferenced detections of slurry tanks that can be imported into GIS.

The primary aim of the model is to develop additional data to inform catchment management and help target catchment walkovers and the location of pollutant control measures. It is thus intended to support existing work rather than replace an existing approach.

1.1.2 Existing Industry Model(s)/Tool(s)

There is no existing model or dataset that shows the locations of slurry tanks. Benchmarking has instead been against a catchment walkover, which would be used to identify a range of potential pollutant issues arising from agriculture, including those associated with intensive livestock farming and the storage of manure (in slurry tanks or other facilities).

1.2 Catchment Selection

Catchment walkovers were undertaken in two catchments (Figure 1 and Figure 2) in different parts of the country.

GB110060029250 (Fernhill Brook - headwaters to tidal limit) in the Carmarthen Bay and the Gower catchment. The area was known to have a number of dairy farms and therefore highly likely to have slurry tanks, bare ground and poached land. The catchment size (11 km^2) was appropriate for a walk over in a day.

GB103024077330 (Smallhope Burn from Source to Browney, Stocke Water Body) – a predominantly grassland catchment with cattle and sheep farms and with the Lanchester Dairy processing and distribution site located within the catchment. The catchment also contains sewage treatment works and equine businesses (with horse walkers), and so may provide challenges for misidentification. The catchment has poor ecological status, in part due to

phosphates and agriculture. The catchment size of 42 km^2 meant that only the lower half of the catchment was covered in detail by the walkover.

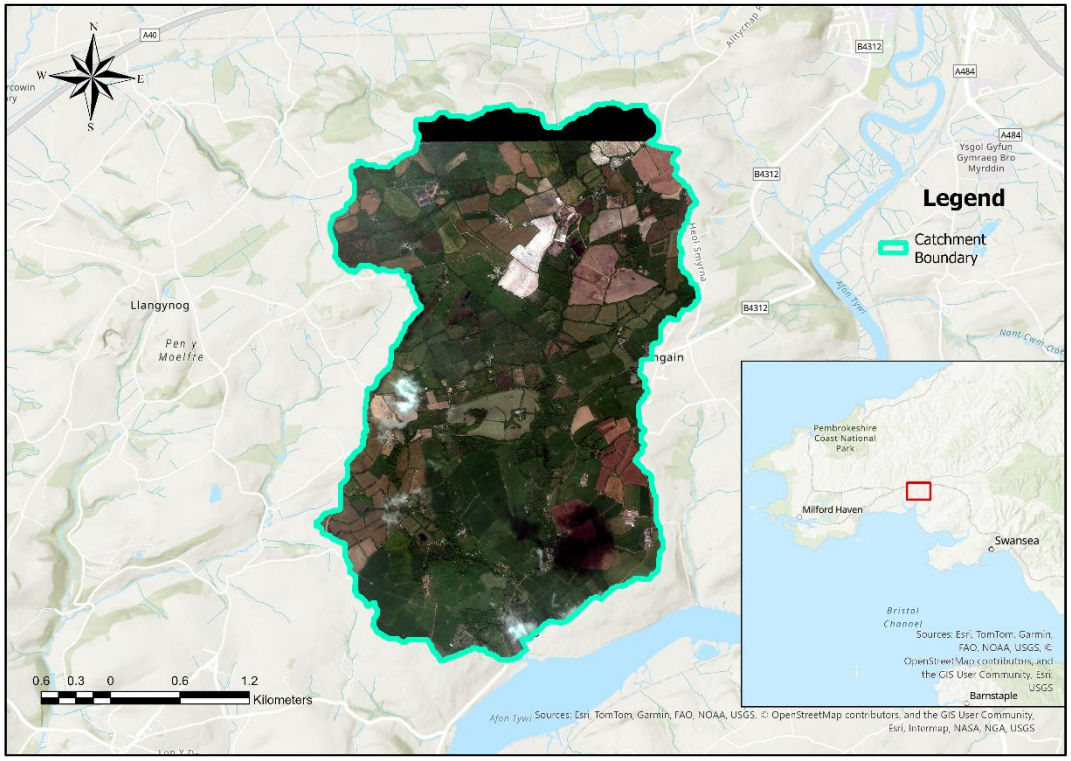

Figure 1 GB110060029250 catchment (Fernhill Brook - headwaters to tidal limit)

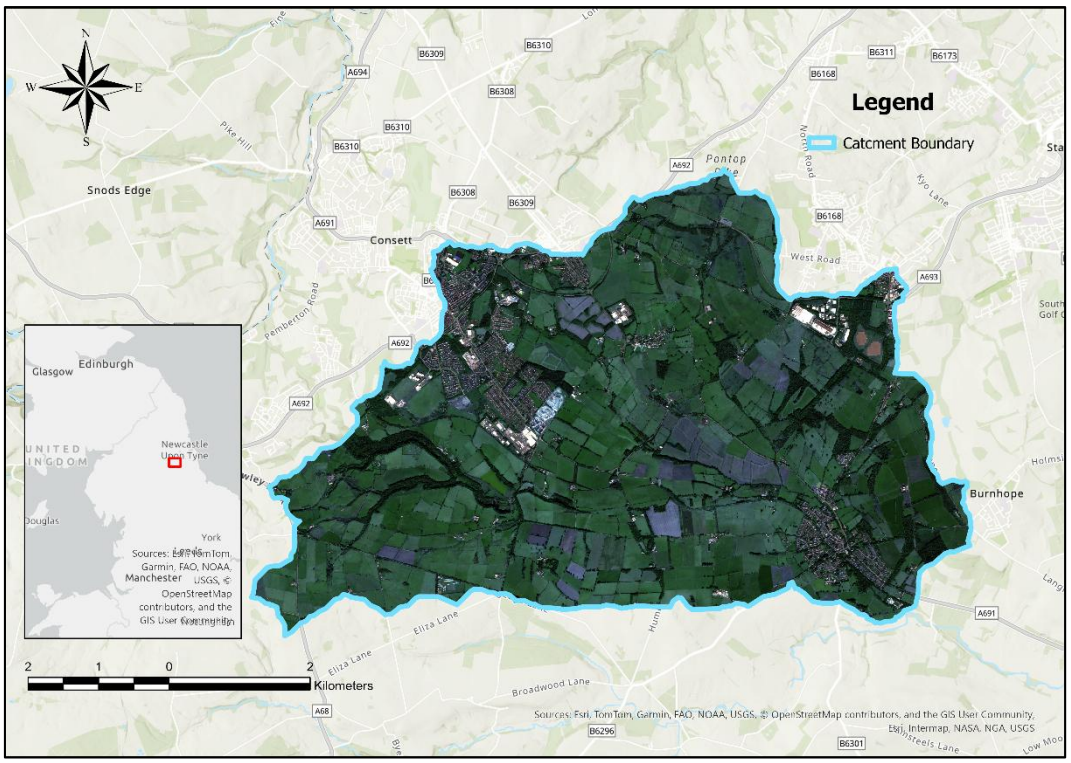

Figure 2 GB103024077330 (Smallhope Burn from Source to Browney, Stocke Water Body)

1.3 Benchmarking Methodology

The benchmarking compares the number and locations of the slurry tanks identified using the model with those identified from the walkovers of the catchments.

Prior to undertaking the actual walkover, the consultant carried out the series of checks that would normally be run through to identify potential locations of interest, high risk areas for diffuse pollution and any accessibility issues. The tools used for this include:

1. Agricultural Land Environmental Risk and Opportunity Tool (ALERT), which contains data layers such as land use, topography and satellite imagery, and was designed to help locate interventions to reduce pollution from agriculture

2. MAGIC in England or DataMapWales, which contain geographic information about the natural environment from across government, to understand the water quality status and pollution issues within the catchments

2. LandIS to identify soils, as pollutant issues can vary by soil type (e.g. lighter texture soils will have a higher nitrate leaching risk than heavier soils, heavier soils are more prone to waterlogging).

4. Google Earth imagery to inspect the catchment and identify areas that should be targeted for inspection (if safe and legal to do so). Date of most recent imagery noted and used, although historic imagery can also be utilised.

Reviewing these datasets for the catchments took several hours, although only item 4 is particularly relevant for identification of slurry tanks.

The walkover was then undertaken, identifying the locations and features of interest, with access constrained to roads, footpaths and bridleways and thus only certain features being visible. The walkover was restricted in time to approximately one day. After the walkover was completed, the information gathered was documented and mapped.

For the application of the RDMAI model, the only significant additional task was the purchasing of the required satellite images that covered the two catchments. Other than that, the model was applied using the files as downloaded from GitHub. Due to issues with different python versions, the slurry model was adapted to run in Google Colab, removing the need to create a virtual Python environment or downgrade to Python 3.10. The high-resolution satellite image and the YOLO model were uploaded to Google Drive, which the adapted Colab Notebook then mounted. From there, it loaded the satellite image and model, and wrote the output back to Google Drive.

1.4 Benchmarking Limitations

Due to access issues and weather (flooding) not all areas could be walked over or assessed. Slurry tanks are not always located in areas visible from public roads and footpaths. However, it is possible to determine the locations of slurry tanks from imagery alone without the need to actually undertake a walkover.

Because slurry tanks are relatively permanent structures (typically with a minimum 20-year designed lifespan), the closeness of the date of any walkover and the satellite imagery are not of any consequence for the benchmarking of this model.

2. Benchmarking Results

2.1 Model 'Ease-of-Use'

Table 2.1 Comparison of model 'ease-of-use'

Model Outputs	Manual review of data / catchment walkover	RDMAI Model
Installation & Run Process	A catchment walkover can be undertaken by anyone with a reasonable understanding of agriculture and diffuse pollution. A more qualified individual, or one familiar with the catchment would speed up the process or provide more detailed insight. Land access is an issue. Not all potential areas of interest are visible from roads, footpaths and other publicly accessible spaces. Someone familiar with the catchment and local landowners may be able to arrange access to areas of private land. These walkovers were restricted to roads, footpaths and bridleways. The catchments visited in this project were approximately 11 km^2 and 42 km^2 in size, and it was not possible to cover the larger catchment in a day, even allowing for some preparatory work to identify areas of interest. The time taken also depends upon the level of detail being recorded. A risk assessment was undertaken prior to the walkover, this included lone working – which whilst avoidable would increase the staffing costs in order to be avoided.	Installation is straightforward if using Python 3.10. A virtual environment, PyTorch, and dependencies from requirements.txt are required. Users should be comfortable with Python, Jupyter Notebooks, and command-line operations. The required model weights file was provided for this work, but is not available for general users who will have to recreate it. This adds additional complexity and uncertainty to the task. The run process is scripted in detection.ipynb and handles tiling, PNG conversion, YOLO inference, and GeoJSON output. Instructions are clear but require attention to Python version and CPU/GPU settings. Overall setup can take 1–2 hours for someone with moderate Python experience.

	Significant rainfall and flooding hindered the walkover of the Welsh catchment, restricting access from certain areas.	
Data inputs	A manual search for slurry tanks can be undertaken using imagery data freely available on the internet such as that included in Google Maps.	Requires high-resolution (sub-1m) satellite or aerial imagery in GeoTIFF format. Images must be georeferenced. Scripts are provided for tiling, PNG conversion, and generating world files, so pre-processing is mostly automated. The user needs to place their imagery in the correct folder structure.
Licensing Requirements	None.	Users must have access to high-resolution imagery. Sentinel-2 or similar free datasets are too low resolution for reliable results. Data at 0.5m resolution was sourced for each catchment at cost of c. £500 per catchment.

2.2 Model Outputs

Table 2.2 Comparison of model outputs

Model Outputs	Manual review of data / catchment walkover	RDMAI Model

Data Format	Outputs were as annotated maps and notes, subsequently converted to GIS shapefiles.	Outputs include PNG tiles of input imagery with YOLO annotations and .txt label files containing normalized bounding boxes and confidence scores. There was also an overview .txt document that listed all the images that found some kind of result. A GeoJSON file was also output that represents location data for where the slurry tanks are.
Temporal Resolution	Representative only of the day of the walkover or the day any imagery used was produced. Whilst more walkovers could be repeated to observe change over time, there are financial and practical constraints given the time required.	The model does not have a built-in temporal component; it runs on the imagery provided. Each run corresponds to the input dataset, so the "temporal resolution" depends entirely on the date of the imagery supplied. In this case the date of the data was 2023.
Geographic Constraints	None. Although land access is an issue, and slurry tanks are frequently located away from publicly available accessible spaces.	The model was trained primarily on UK imagery (including Wales and England) as well as Dansih datasets. It should run anywhere with similar spatial resolution and spectral characteristics (RGB bands, 0.3–1 m pixel resolution). Limitations arise if imagery differs significantly in resolution, spectral bands, or landscape features, as this can reduce detection accuracy. As a test, the model was run on lower resolution data (10m) and this yielded very poor results. Licensing restrictions may also apply for certain imagery sources, e.g., ESRI basemaps or commercial high-res satellite data.

2.3 Time and Cost Comparison

A fairly rapid catchment walkover for a small (~10 km2) catchment can be undertaken in one day, plus some additional time for preparation (e.g. reviewing datasets) and reporting. However, it must be noted that identification of slurry tanks could be undertaken without a walkover using imagery data freely available on the internet such as that included in Google Maps – this was done in around 1 hour for the Northumbrian catchment (GB103024077330), with most of that time spent viewing imagery for one particular site (which it was eventually concluded was a horse walking area). Unfortunately, this site was only recently constructed and so was not in the data from 2023 used for the RDMAI model - thus the model's interpretation of this site is unknown.

The RDMAI model potentially offers time savings compared to manual identification, particularly if a walkover (taking ~1 day) is undertaken;. However, the cost savings depend heavily on the availability of suitable high-resolution imagery for said study area.

The model requires imagery with a resolution of ≤1 m, which must be purchased commercially. Obtaining suitable imagery is therefore the largest cost factor and may outweigh the time savings gained from automated processing. Once appropriate imagery is obtained, the model run time is very fast (typically minutes for a 25 km² area), and the workflow can be repeated with minimal additional effort.

However, the model requires a moderate level of technical expertise, including Python environment setup, package version management, and basic image pre-processing.

Overall, the RDMAI model would reduce labour time spent on surveying large areas for potential slurry tanks, but still requires manual verification due to false positives. The overall cost efficiency would therefore depend on the cost of imagery, and any time for manual verification.

In reality however, the slurry tank identification model would not be used in isolation, and would form part of a toolset (alongside other models from RDMAI and the data listed in Section 1.3), to help information any catchment walkovers undertaken, rather than as an alternative to catchment walkovers.

2.4 Model Performance

Figure 3 shows that one slurry tank was found in the Welsh catchment as part of the walkover, whilst the RDMAI model identified two (using a confidence threshold of 0.8). The satellite imagery for the slurry tank found from the walkover is shown in Figure 4 – the area between the tank and the neighbouring building is not particularly clear, and there is no distinct round feature. The satellite imagery from which the RDMAI model detected the tanks is shown in Figure 5 and Figure 6. Further investigations of the area shown in Figure 6 revealed that this was not a slurry tank but Pantyrathro wastewater treatment works (WWTW). Because of their potential for confusion, WWTW works were specifically included in the training and testing of the model, although they may not have had the single filtration bed found here.

Note that the default model script outputs anything over the confidence threshold of 0.5, yet even with value of 0.7, the model was still showing numerous false positives, primarily of large round trees, some examples of which are shown in Figure 7.

Despite the presence of large numbers of cattle and the Lanchester Dairies processing site in the catchment, neither the model nor the walkover found any slurry tanks in the Northumbrian catchment. Multiple sewage treatment works were identified (with a confidence value of around 0.75). However, numerous other alternative types of manure storage were identified as part of the walkover (Figure 8), reinforcing the need to expand the RDMAI model for it to have greater utility.

It is probably appropriate to use a threshold value of 0.8 in order to exclude the most obvious false positive errors, but a larger dataset would be necessary to make this a firm conclusion.

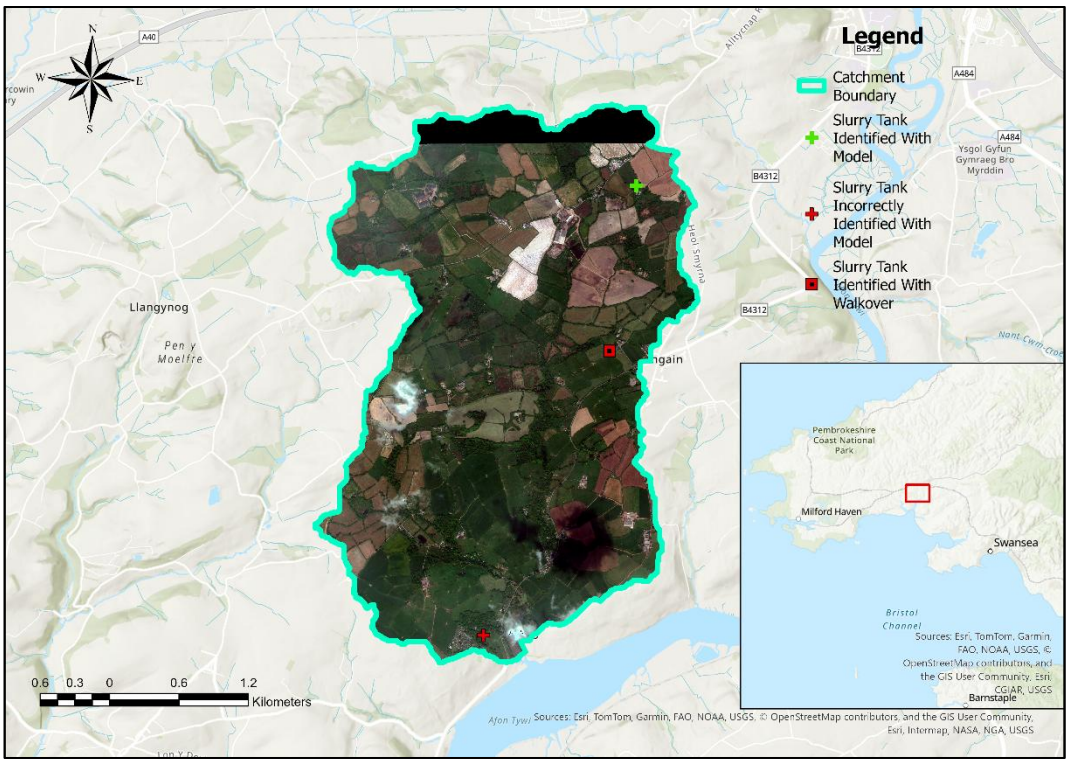

Figure 3 Locations of the slurry tanks in the Welsh catchment identified by the RDMAI model and the catchment walkover

Figure 4 The slurry tank identified as part of the walkover of the Welsh catchment, but not detected by the RDMAI model

Figure 5 The slurry tank detected in the Welsh catchment by the RDMAI model, but not identified as part of the walkover.

Figure 6 The slurry tank incorrectly identified by the RDMAI model. The feature is actually Pantyrathro WWTW.

Figure 7 Features incorrectly identified as slurry tanks in the Welsh catchment when using a slightly lower (0.7) confidence threshold for the RDMAI model

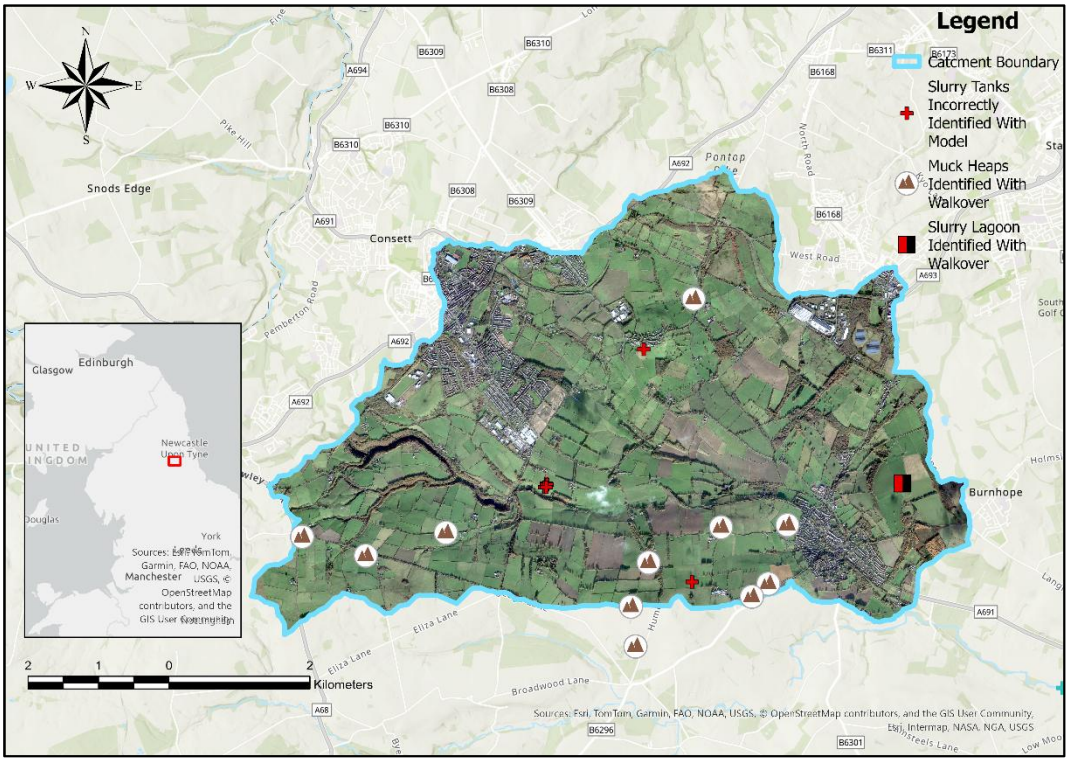

Figure 8 No slurry tanks were identified in the Northumbrian catchment by the RDMAI model or the catchment walkover. The image shows the locations of mis-identified slurry tanks from the RDMAI model (using a 0.7 threshold) and alternative manure storage features found through the catchment walkover.

3. Benchmarking Conclusion

3.1 Summary

A summary table of key model characteristics can be found in Table 1.1, at the start of the report.

3.2 Conclusion

The RDMAI slurry tank detection model offers a potentially quicker automated alternative to manual identification methods, with time savings when suitable high-resolution data is available. However, several limitations currently restrict its practical use.

- The model requires sub 1 metre aerial resolution imagery, which is costly (c. £10 per km^2). Free data like sentinel-2 are too low-resolution.

- Installation was complicated by Python and PyTorch version constraints, requiring a manual downgrading of the software currently installed. An alternative approach was found where the slurry model notebook was adapted to run in Google Colab, removing the need to create a virtual Python environment or downgrade to Python 3.10. The high-resolution satellite image and the YOLO model were uploaded to Google Drive, which the adapted Colab Notebook then mounted. From there, it loaded the satellite image and model, and wrote the output back to Google Drive

- Whilst ADAS were provided with the 'model weights' file, this is not available in the Github file due to licensing constraints, so additional imagery and processing would be needed to create this the first time an organisation were to use the model.

- A manual review of freely available imagery took comparable time to running the model, so time savings would only be realised if the model was applied at scale, by someone familiar with it.

Although the purchasing of satellite imagery for the slurry tank model is expensive when considered in isolation, such imagery could potentially be used for other applications once purchased, including some of the other RDMAI models (subject to modification of their code).

Model performance during testing showed sensitivity to image quality, with many false positives at confidence thresholds below 0.8, picking up features such as trees and sewage treatment works. However, it did find a tank that was missed as part of the walkover.

The model currently only identifies above ground circular slurry tanks, which only account for around a quarter of manure storage facilities – slurry can also be stored in alternatively designed tanks or lagoons, and a significant proportion of manure is managed as solid waste and so is just stored in manure heaps (as shown by the Northumbrian walkover). If the model could be

expanded to identify these other manure management facilities, and additional potential point sources such as silage clamps, it would become significantly more useful.

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